

Math 110 – Homework 6 Answers

Math 110 – Fall 2026

Homework 6

Invariant subspaces and eigenvalues

Suggested completion date: Friday, October 9

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Let $P : V \rightarrow V$ satisfy $P^2 = P$.

- (a) Prove that every eigenvalue of P belongs to $\{0, 1\}$.
- (b) Prove that $V = \text{null } P \oplus \text{range } P$.

Problem 2. Let $D : \mathcal{P}_n(\mathbb{F}) \rightarrow \mathcal{P}_n(\mathbb{F})$ be differentiation, where $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$.

- (a) Determine all eigenvalues and eigenspaces of D .
- (b) Explain why the answer is different if one replaces $\mathcal{P}_n$ by the space of smooth functions on $\mathbb{R}$.

Problem 3. Let $S : V \rightarrow V$ be invertible and set $R = S^{-1}TS$. Prove that T and R have the same eigenvalues. More precisely, prove that

$$E(\lambda, R) = S^{-1}E(\lambda, T)$$

for every scalar λ .

Problem 4. Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Using only the 2×2 determinant formula from Homework 4, prove that

$$\det(tI - A) = t^2 - (a + d)t + (ad - bc).$$

Then prove that a scalar λ is an eigenvalue of A if and only if it is a root of this polynomial.

Problem 5. For each of the following matrices, find the eigenvalues and eigenspaces over the indicated field, and decide whether the matrix has a basis of eigenvectors:

$$A_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

For A_1 , work first over $\mathbb{R}$ and then over $\mathbb{C}$; for A_2 and A_3 , work over $\mathbb{R}$.

Problem 6 (optional). Let $T : V \rightarrow V$ and let $v \neq 0$. Define the cyclic subspace

$$\mathcal{C}(v, T) = \text{span}(v, Tv, T^2v, \dots).$$

- (a) Prove that $\mathcal{C}(v, T)$ is T -invariant.
- (b) If V is finite-dimensional, prove that there is some $m \leq \dim V$ such that

$$\mathcal{C}(v, T) = \text{span}(v, Tv, \dots, T^{m-1}v).$$

Problem 7 (optional). Consider the column-stochastic matrix

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{pmatrix}.$$

Find a probability vector p satisfying $Pp = p$, and explain its interpretation for the corresponding three-state Markov chain.

Solutions

Problem 1

Solution notes

Problem 2

Solution notes

Problem 3

Solution notes

Problem 4

Solution notes

Problem 5

Solution notes

Problem 6

Solution notes

Problem 7

Solution notes
